



COMPLEX ENGINEERING PROBLEM (CEP)

Artificial Intelligence (CSC-411)
Spring 2026

Instructor	<u>Engr. Saad Mazhar Khan</u>
Section	<u>(A, B)</u>
Group Members	<u>3 Members Allowed</u>

General Instructions

1. Students must complete this Complex Engineering Problem (CEP) in groups of 2–3 members.
2. The selected problem domain must be real-world and clearly justified.
3. The project must demonstrate the application of Artificial Intelligence techniques, particularly problem formulation, search strategies, and heuristic design.
4. All work submitted must be original. Any form of plagiarism will result in disciplinary action as per university policy.
5. Students are encouraged to use reliable academic sources, tools, and datasets to support their work.
6. Plagiarism will be treated strictly, and any form of academic misconduct will be dealt with in accordance with university rules and regulations.

1. Introduction

This CEP evaluates students' ability to apply Artificial Intelligence techniques to real-world problems by integrating search, reasoning, and decision-making capabilities into a system.

2. Problem Statement

Students must design an AI-based system that solves a real-world problem (e.g., disaster response or smart traffic) by modeling the environment and applying intelligent algorithms.

3. **Assignment 3: AI System Design & Implementation (05-05-2026)**

Students are required to implement the core AI solution by defining the problem, designing the state space, and applying search algorithms to simulate intelligent decision-making.

- a) **Problem Formulation:** Define initial state, goal state, actions, and transitions clearly to structure the AI problem logically.
- b) **State Space Representation:** Model the environment as a grid or graph to visualize how the agent explores solutions.
- c) **BFS Implementation:** Apply Breadth-First Search to find optimal paths and demonstrate its working with examples.
- d) **A* and Heuristic Design:** Design an admissible heuristic and use A* to improve efficiency over BFS.
- e) **CRUD & Testing:** Implement basic operations and test system behavior to ensure correctness.

4. **Assignment 4: Documentation & Evaluation (19-05-2026)**

Students must document the system comprehensively, explaining design decisions, AI methods, and evaluating performance.

- a) **System Architecture:** Present a clear diagram showing interaction between components.
- b) **AI Methodology:** Explain algorithms used and justify their selection.
- c) **Testing & Results:** Provide results demonstrating system correctness and efficiency.
- d) **IEEE Paper:** Prepare a professional 6-page paper summarizing the entire work.

5. Washington Accord Mapping

WK – Knowledge Profile

Code	Description	Mapping Explanation
WK1	Understanding of computing systems	Involves modeling the AI problem using states, transitions, and agent-based design, forming the foundation of intelligent systems.
WK3	Knowledge of mathematics, statistics, and AI	Requires designing admissible and consistent heuristics, applying AI concepts, and mathematical reasoning.
WK4	Engineering fundamentals	Involves algorithm design and analysis to solve real-world problems under constraints.
WK6	Knowledge of tools and technologies	Applies AI techniques such as BFS and A*, along with simulation and implementation tools.

WP – Problem Solving Profile

Code	Description	Mapping Explanation
WP1	Problem Identification	Translating real-world scenarios into a structured AI problem formulation.
WP2	Problem Analysis	Evaluating different search strategies and selecting appropriate approaches.
WP3	Design and Development	Developing AI-based solutions including state space, algorithms, and heuristics.
WP5	Multi-disciplinary Integration	Integrating environmental factors, data, and computational models into a system.
WP6	Engineering Judgment	Selecting suitable AI techniques and justifying decisions (e.g., A* vs BFS).

6. Evaluation Rubric

Criteria	Marks
Problem Formulation	10
State Space Modeling	10
Search Implementation	20
Heuristic Design	15
System Design	15
Testing & Evaluation	10
IEEE Paper	10
Viva	10

7. Concluding Remarks

This Complex Engineering Problem (CEP) is designed to evaluate students' ability to apply Artificial Intelligence concepts to real-world scenarios through systematic problem formulation, algorithm design, and solution development. Students are expected to demonstrate not only technical implementation but also critical thinking, analytical reasoning, and engineering judgment throughout the project.

The emphasis of this CEP is on developing a well-structured, intelligent system that integrates appropriate AI techniques with clear justification of design choices. Students must ensure that their solutions are efficient, scalable, and logically sound, supported by proper testing and evaluation.

Successful completion of this CEP will reflect the students' readiness to tackle complex engineering problems and their ability to bridge theoretical knowledge with practical application in the field of Artificial Intelligence.